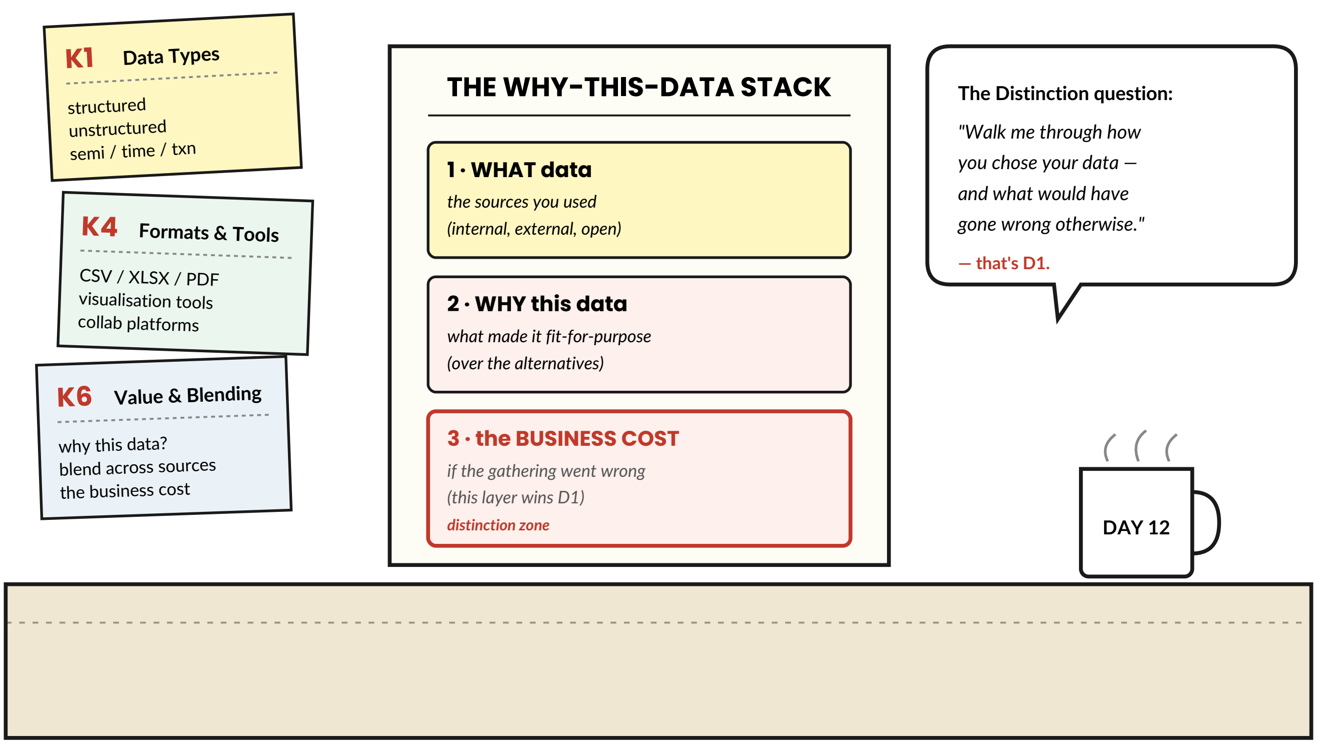
12

first content area · the why-this-data day

Data Gathering, STARR-ed

K1 · K4 · K6 · and the D1 distinction setup



Three KSBs, one STARR shape, one new mini-device — **The "Why This Data?" Stack**. Today's chapter takes your data-gathering anchor stories and turns them into the kind of answer that defends at Distinction.

What the assessor will ask

Three KSBs sit inside Data Gathering. Each has a typical question shape — below: what each KSB asks for, what good answers look like, and where the distinction lives.

THE THREE DATA-GATHERING KSBs

K1
Data Types

five types you'll know

· **Structured** — tables, schemas
· **Unstructured** — text, image
· **Semi-structured** — JSON/XML
· **Time-series** — sensor logs
· **Transactional** — events

Q shapes

"What kinds of data do you work with?"

"Give me an example of unstructured data."

your anchor: name one example of each

K4
Formats & Tools

three things at once

· **Data formats** for analysis
(CSV, XLSX, JSON, PDF)

· **Visualisation tools**
(Power BI, Tableau, Excel)

· **Collab tools**
(Teams, SharePoint, Git)

Q shapes

"Which tools did you use, and why those?"

"What format made the analysis easier?"

link tool choice to the question being answered

K6
Value & Blending

two-part KSB

· **Value to the business**
why is this data worth gathering at all?

· **Blending sources**
combining internal + external for richer view

Q shapes

"Why does data matter to your organisation?"

"Describe a time you blended two datasets."

K6 is the D1 setup — deepest answers here

The "Why This Data?" Stack

A three-layer extension that lifts a data-gathering Reflection toward D1. You add it inside the second R of STARR. If you only remember one thing from this chapter, remember the third layer — *"what would have gone wrong otherwise."*

1 WHAT

the data you used

"Epic activity feed as primary; provider invoice as cross-check."

→

2 WHY

why this, not the others

"Epic was clinical truth; invoice was what we'd been charged for."

→

3 COST

what would have gone wrong

"If I'd used invoice as truth, we'd have rubber-stamped billing errors."

SHOW THIS OFF

Most apprentices nail layer 1 and miss layers 2 and 3. The assessor is looking for all three — especially the third, which is what D1 literally asks for: *"evaluates how incorrect data gathering can affect the output."* Layer 3 is the answer to that descriptor.

A full STARR with The "Why This Data?" Stack inside

Below: a Data-Gathering question answered in STARR shape, with the three-layer Why Stack folded into the Reflection. Total time: ~70 seconds.

A DATA-GATHERING STARR — 70-second transcript

Q: "Walk me through a time you had to choose between different data sources to answer a business question."

S — **SITUATION**
"In May, our finance lead asked for a monthly externally-provided activity report. Two possible sources."

T — **TASK**
"I had to pick between the provider's own invoice extract and our internal Epic activity feed."

A — **ACTION**
"I went with the Epic feed as primary — structured, NHS-number-keyed, refreshed nightly. I used the provider's invoice as a verification cross-check. The reason: the Epic data was my source of clinical truth; the invoice told me what we'd been charged but not whether the procedure had actually happened."

R — **RESULT**
"The report flagged 14 invoice/activity mismatches a month. About a third were legitimate billing errors."

R — **REFLECTION — the D1 layer: evaluate & justify the choice, and the cost of getting it wrong**
"On reflection, the choice mattered more than the analysis. If I'd taken the invoice as the source of truth instead, we'd have rubber-stamped the billing errors — the very thing we were trying to catch. The lesson: when you're verifying a third party's data, never let their data be your sole source. The choice between sources isn't arbitrary; it's the verification itself. I'd take that into any future reconciliation work."
↑ this is D1: evaluates the data-set choice, justifies it, names what would have gone wrong otherwise.

D1 in plain English

D1 is the first of three Distinction descriptors. It applies to the Data Gathering content area. The official wording is dense; the plain-English version is shorter:

The official wording

> *"Evaluate and justify why using different data sets is important to the business, and evaluate how incorrect data gathering can affect the output."*

In plain English

Two moves:

1. Justify *why this data* rather than something else.
2. Name *what would have gone wrong* if you'd picked badly.

The Why Stack delivers both moves in one go.

Pass-level → Distinction-level: side-by-side

Pass-level answer	Distinction-level answer
"I used the Epic feed for the report because it was easier to access."	*"I used the Epic feed because it was the source of clinical truth. The provider invoice was secondary — using it as primary would have meant rubber-stamping the billing errors I was trying to catch."*

ASSESSOR FAVOURITE

The phrase *"what would have gone wrong otherwise..."* is the D1 trigger. Use it in at least one Data Gathering answer in the discussion. Even if you cover the first two Why-Stack layers naturally, missing the third costs distinction marks every time.

What they're likely to ask

The data-gathering portion of the discussion typically opens with one or two of these. Rehearse a STARR-shaped answer to each — using your anchors, not StreamWave examples.

1. "What kinds of data do you work with day-to-day?"

Shape: a quick STARR-Lite tour through three of the five data types, with one workplace example each. Don't try to cover all five.

2. "Walk me through a time you had to pick between different data sources."

Shape: full STARR. Reflection layer must include the Why Stack — especially the third layer (what would have gone wrong).

3. "Describe a piece of work where you blended two or more data sources."

Shape: full STARR. The Action layer is where the technical detail of the blend lives (matching keys, identifying overlaps, handling mismatches). The Reflection links to *why blending mattered* for the answer.

4. "Why does data matter to your organisation?"

Shape: a short opening on context, then a specific example of a decision data drove. Tie K6 to a real moment, not a slogan.

5. "What tool did you use and why those tools?"

Shape: K4. Quick mention of tools, longer mention of *why those tools fit the question*. Don't list features; link tool to purpose.

SHARPEN YOUR PENCIL

30 minutes — Why-Stack your data-gathering anchor

Pick your data-gathering anchor story (from Day 10). For each layer of the Why Stack, write three or four words. The full sentence comes when you say it out loud.

My data-gathering anchor: *name it here:*

Layer 1 — *WHAT data did you use?*

Layer 2 — *WHY this data and not another?*

Layer 3 — *what would have gone WRONG with a different choice?* *(the D1 layer)*

Halfway House

Five quick questions. Answer in your head first; check below.

1. The Why-This-Data Stack lives inside the Reflection layer of STARR.

☐ **True** ☐ **False**

2. Layer 3 of the Why Stack ("what would have gone wrong") is optional.

☐ **True** ☐ **False**

3. D1 asks you to evaluate the choice of data sets and the cost of getting that choice wrong.

☐ **True** ☐ **False**

4. When asked about K1, I should try to cover all five data types with workplace examples.

☐ **True** ☐ **False**

5. K6 is the KSB most aligned with D1 — the deepest answers belong here.

☐ **True** ☐ **False**

Answers

1. True. The Why Stack *is* the depth move you add inside Reflection.

2. False. Layer 3 is the layer that wins D1. Skipping it is the most common reason apprentices miss Distinction here.

3. True. Two moves: justify the choice, name the cost of the alternative.

4. False. Three types with one workplace example each is stronger than five types covered briefly.

5. True. K6 (value + blending) is where the Why Stack lands most naturally and where D1 is graded.

There are no *Dumb* Questions

Q: *My anchor story uses only one data source. Does that still work for K6 / blending?*

A: For K6 you'll want a different anchor — or, in the same story, name the source you considered and chose not to use. *"I used the Epic feed; I considered the manual provider extract and decided against it because..."* That's enough to evidence blending-thinking even if you didn't blend in the end.

Q: *I struggle to remember what counts as semi-structured data.*

A: Two examples are enough — JSON or XML files, and email headers (structured envelope, free-text body). If asked, give one of those and move on.

Q: *What if my Layer-3 "what would have gone wrong" feels too theoretical — it didn't actually happen?*

A: That's exactly what D1 wants: *"evaluates how incorrect data gathering can affect the output."* The descriptor is about reasoning, not about an actual incident. Pose the alternative scenario and walk through what it would have produced.

Q: *The assessor asked about tool choice but I just used Excel for everything. Is that a problem?*

A: No — provided you can justify why Excel fit. *"I used Excel because everyone on the team can open it; Power BI would have been more powerful but harder to share."* Tool choice is about fit, not sophistication.

Q: *I work in a small team and we don't really "blend" datasets formally. What do I say?*

A: Almost everyone blends without naming it. If you've ever VLOOKUP-ed one sheet against another, cross-referenced a spreadsheet against an email, or built a report drawing on two systems — that's blending. Tell that story.

Say it for real

Two questions to rehearse out loud. Use your anchors; finish each with the Why Stack in the Reflection layer.

Q1 — Walk me through a time you chose between different data sources.

The shape: full STARR. Reflection ends with all three Why-Stack layers — especially *"what would have gone wrong otherwise"*.

Try: end the answer with: *"...if I'd done it the other way, [name the failure mode]. The lesson I'd take into future work is [generalised principle]."*

Q2 — Describe a piece of work where you blended data from multiple sources.

The shape: full STARR. The Action layer carries the technical detail (matching key, dedup, fan-out check); the Reflection layer carries why the blend was the right move and what would have gone wrong without it.

SHOW THIS OFF

Land the phrase *"the choice between sources isn't arbitrary; it's the verification itself"* — or some version of it — in your Reflection. It's the kind of generalisation that signals you've thought past the technique into the principle. Distinction-level move.

Bullet Points — the chapter in 30 seconds

- **Three Data-Gathering KSBs.** K1 (data types) · K4 (formats & tools) · K6 (value & blending).
- **The Why-This-Data Stack.** Three layers: WHAT · WHY · what would have gone WRONG.
- **The Why Stack lives inside Reflection.** Not as its own paragraph — folded into the second R of STARR.
- **Layer 3 wins D1.** "What would have gone wrong otherwise" is the literal D1 trigger.
- **Tool choice is about fit, not sophistication.** Justify why the tool matched the question.
- **K6 is the depth KSB.** Expect the longest, most reflective answers here.

TEST DRIVE

Tick each only when it's true:

- [ ] You can name the three layers of the Why-This-Data Stack from memory.
- [ ] You've rehearsed at least one full STARR with all three Why-Stack layers folded into Reflection.
- [ ] You can give one workplace example of each: structured, unstructured, semi-structured data.
- [ ] You can articulate *why* you chose your tool of choice in your anchor story.
- [ ] You've practised the phrase *"what would have gone wrong otherwise…"* out loud.

COMING UP — DAY 13

Analysis & Validation, STARR-ed. Tomorrow we take the second content area — K10 (data quality issues), K11 (validation methods), K15 (role of data in the digital world) — and walk them through STARR. Named device: **The Verification Story** — a four-beat shape (issue found · method used · corrective action · verified) that pulls Phase 1's Two-Route Check and Three Questions Before You Delete into the AM2 conversation.

that's Day 12 done — data gathering, rehearsed.